Small Scale Air-Sea Interaction Problem Set 1 Writeup

Kade Jimenez

Data Cleaning:

The instructions stated that it would be ideal to only use best quality data, which corresponds to the "Sbytes" variable being tagged as 0. After combining all of the data sources, it was noted that 15.82% of the data (2019 out of 12760 measurements) did not fit this criteria and was discarded.

Buoy Location:

The OOI Pioneer buoy array sits off the coast of New England at the boundary between cold continental shelf water and warmer, saltier water associated with the Gulf Stream. The location is also on a mid-latitude storm track, meaning that during the winter season large spikes in measurements will likely be seen.

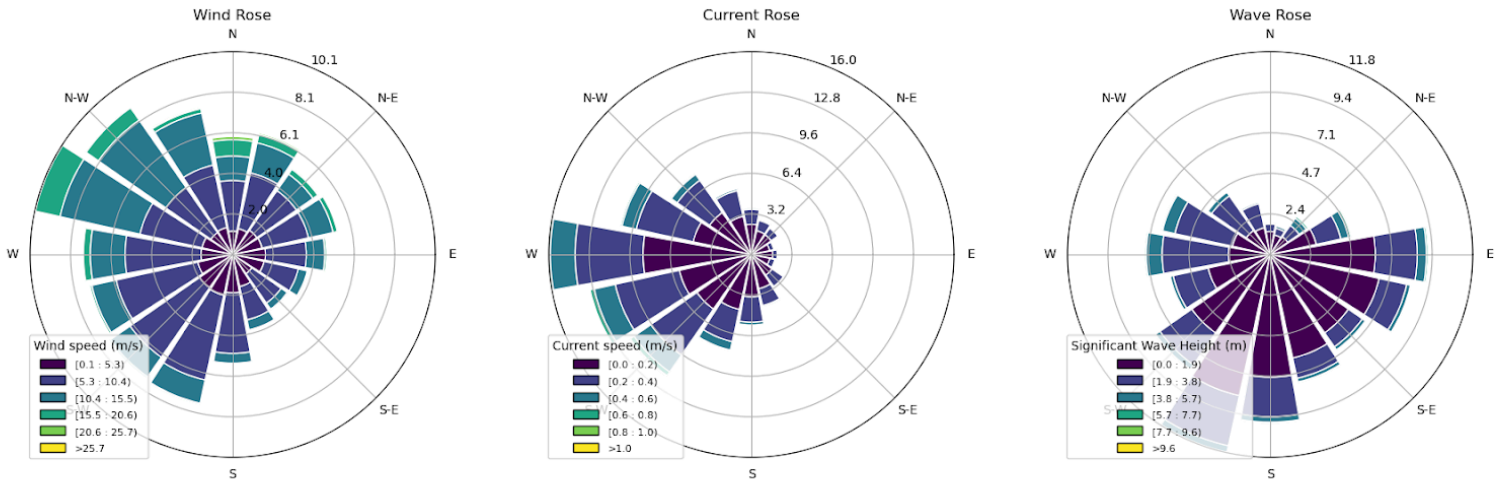
Task #1:

1. The variable names for the direct covariance fluxes are depicted in the table below

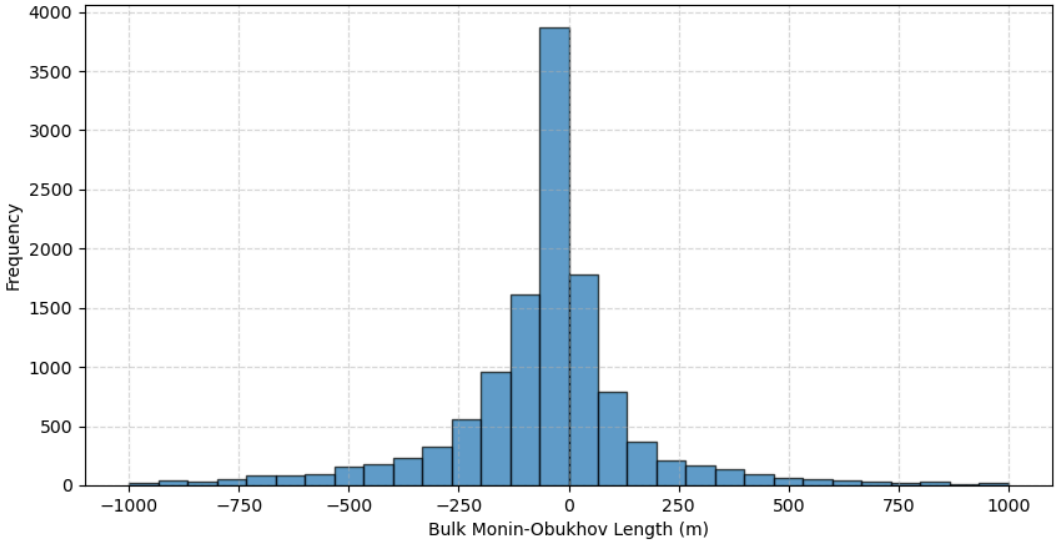
Measurement	Variable Name
Advective flux relative to water	UVr
Wind component of stress relative to water	Along wind: UWr Cross wind: VWr
Wind component of stress relative to earth	Along wind: UW Cross wind: VW
Advective flux relative to earth	UV
Sonic buoyancy flux relative to water	Vertical: WTr Along wind: UTr Cross wind: VTr
Sonic buoyancy flux relative to earth	Vertical: WT Along wind: UT Cross wind: VT

2. The rose plots below illustrate the magnitude and direction of wind, current, and waves. The roses follow the meteorological convention, indicating the direction the phenomena originate from. While there is some directional variability, the strongest winds originate from the WNW, consistent with the prevailing mid-latitude westerlies. Currents display a strongly unimodal distribution, primarily originating from the west, with a maximum

velocity of 0.69 m/s. This dominant direction likely reflects persistent indicates along-shelf flow is the dominant driver over wind forcing. Significant wave heights show a somewhat bimodal directional distribution, with peaks in the SSW and E directions and a maximum value of 9.58 m. The larger SSW peak can likely be attributed to long fetch swell propagating in from distant Atlantic storms, while the eastern peak may reflect storm-generated waves from regional storms such as Nor'easter. The cluster of smaller waves originating from the W and WNW are likely fetch-limited, locally generated wind seas.



3. A scatter plot of Monin-Obukhov length (L) from the data is shown below. Several notable outliers are present, so values outside the bounds -1 km < L < 1 km were excluded, resulting in a plot that still contains 95% of the data. Since these values are bulk estimates calculated using COARE 3.5, some combination of sensor inaccuracies and near-zero denominators likely contributed to the extreme outliers.

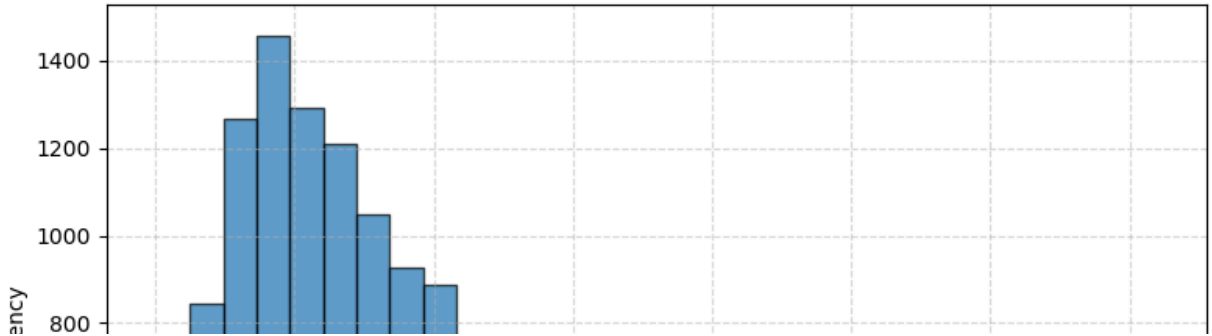


To minimize the influence of extreme values while still considering the full dataset, the measurements were categorized into three stability regimes:

Regime	Definition	Percentage of Measurement (%)
Unstable	L < -50 m	42.67
Near-Neutral	-50 m < L < 50 m	36.71
Stable	L > 50 m	20.62

The distribution of L is unimodal, with the majority of observations between –250 m and 250 m. While only about a third of measurements fall in the strict near-neutral range (–50 m < L < 50 m), the bulk of the data indicate that mechanical turbulence from wind shear is the dominant source of atmospheric mixing in this region. This pattern is likely influenced by the site's location in the storm track and the persistent mid-latitude westerlies.

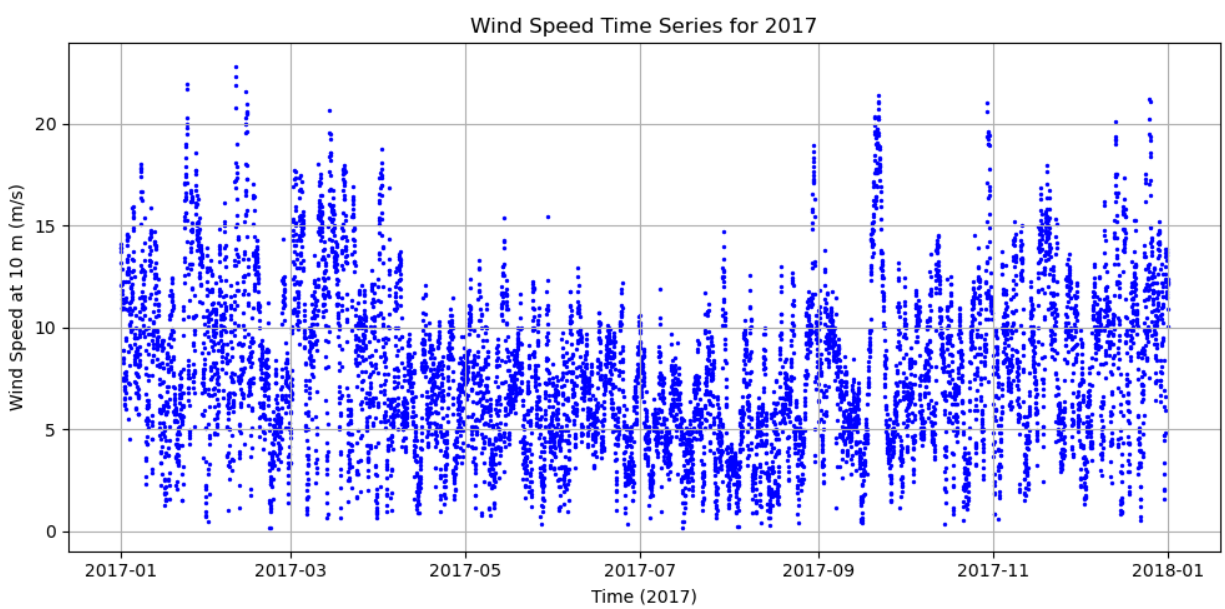
4. A histogram of the gustiness factor across the entire sampling period is shown below. The gustiness factor is defined as gf=S/U where S is the wind speed and U is the magnitude of the mean wind vector obtained from vector averaging. Similar to the Monin-Obukhov length analysis, we divide the gustiness factor into regimes:



Regime	Definition	Percentage of Measurement (%)
Steady	$0 < gf < 0.2$	31.13
Moderate gusts	$0.2 < gf < 0.4$	35.95
Strong gusts	$0.4 < gf < 0.6$	19.89
Very strong gusts	$0.6 < gf$	13.03

These values indicate that wind at the buoy site is generally mild to moderately gusty, with over 65% of observations having a gustiness factor of 0.4 or less. Nevertheless, a substantial fraction of measurements (~33%) fall into the strong or very strong gust categories, reflecting the influence of the storm-track environment and persistent westerly winds in the region.

5. To assess the influence of the diurnal cycle on SST, we examined the mean wind speeds at 10 m for 2017. Winds are relatively strong in winter (≈10 m/s) and weaker in summer (≈5 m/s). Because strong winds enhance turbulent mixing, the SST diurnal cycle is likely less pronounced during winter months. In contrast, during summer, the lower wind speeds and longer days allow the surface layer to warm more during the day, so diurnal variations are expected to be most pronounced during these periods, particularly on calmer, low-gust days.



Task #2

See the Jupyter Notebook containing all the code for examples of my use of AirSeaFluxCode.

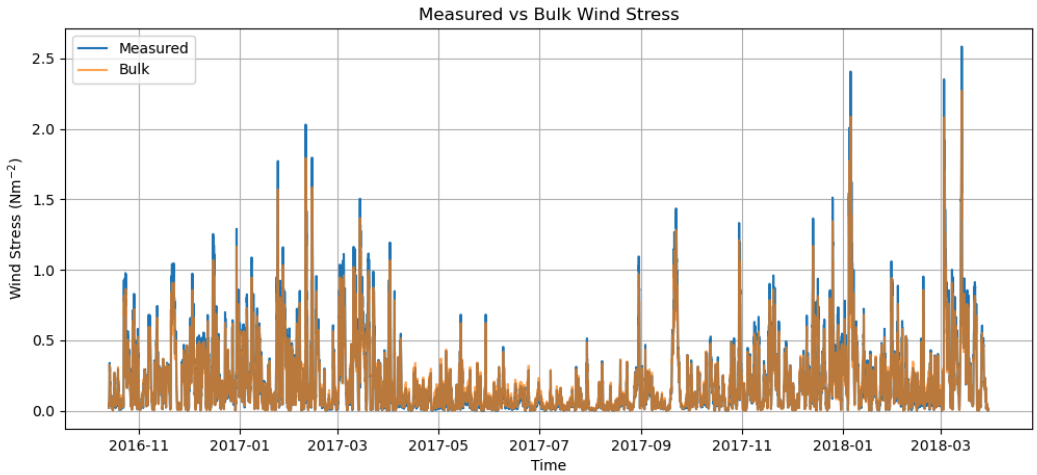
Task #3

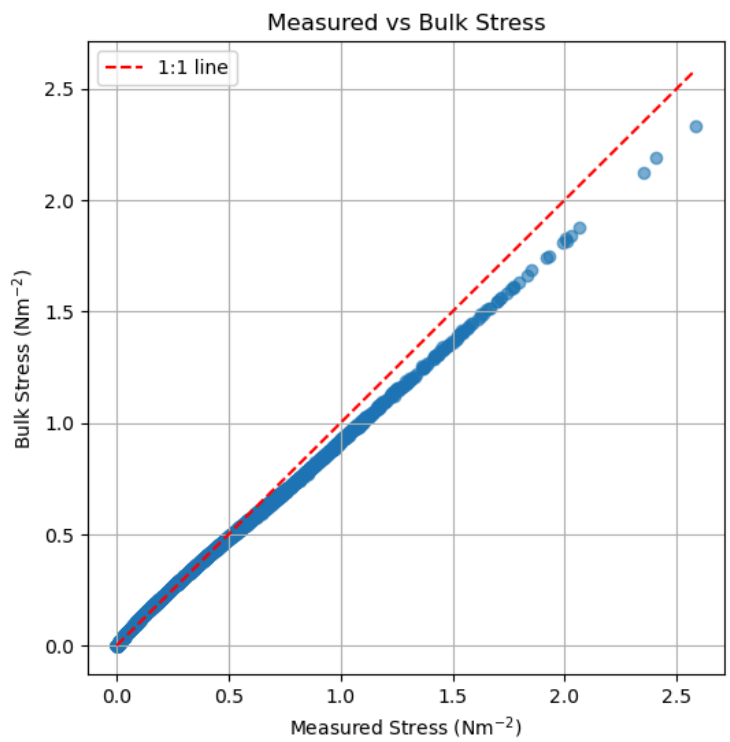
After using AirSeaFluxCode to estimate wind stress, a small fraction of the dataset (~0.8%) was flagged for missing or unphysical values, such as relative humidity exceeding 100%. Given the very small proportion of affected measurements, these data points were removed from further analysis.

An initial examination of the parameterized wind stress, drag coefficient (Cd), and roughness length (z0) indicated that wind stress is the most appropriate variable for detailed analysis, as it exhibited the fewest outliers and is directly comparable to measured momentum flux. While Cd and z0 are fundamental components of the parameterization, wind stress is the physically meaningful quantity governing air-sea momentum transfer, making it the clearest metric for evaluating scheme performance.

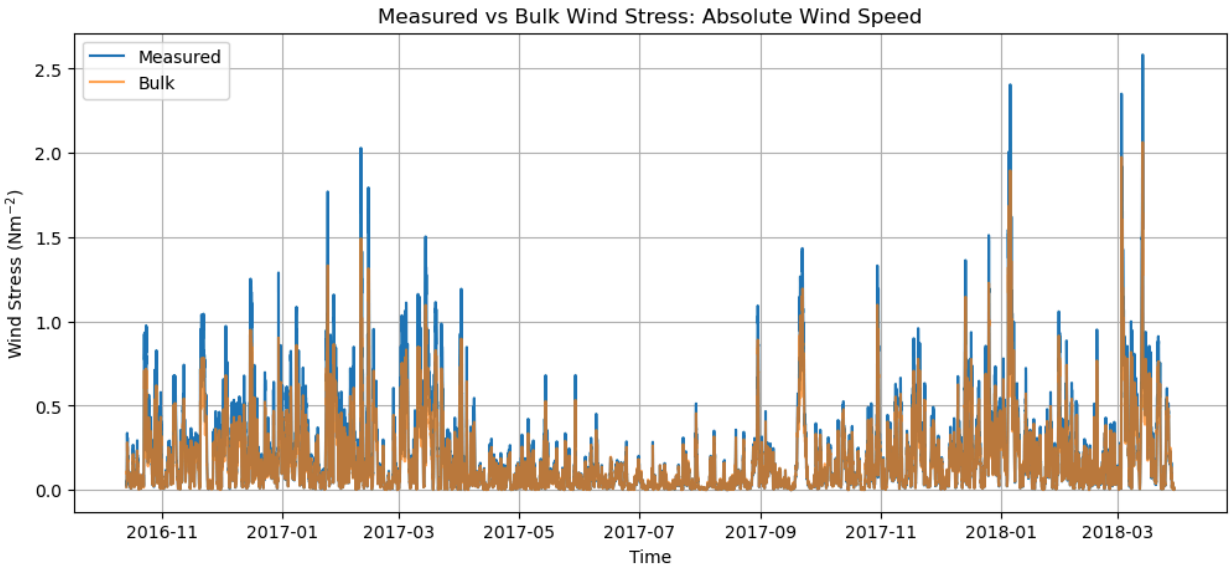
The selection of a bulk flux parameterization considered the occurrence of high-wind events, with measured winds exceeding 20 m/s during several storm events. Parameterizations with limited wind speed ranges were excluded, and some schemes, such as LP82, failed to converge for certain data points. Ultimately, ECMWF was chosen, as it incorporates wave-dependent roughness and drag coefficients, providing realistic estimates across the observed wind range. It should be noted that the ECMWF scheme caps drag coefficients at very high winds, which can slightly underestimate fluxes during extreme events; however, the scheme overall matches the measured data closely.

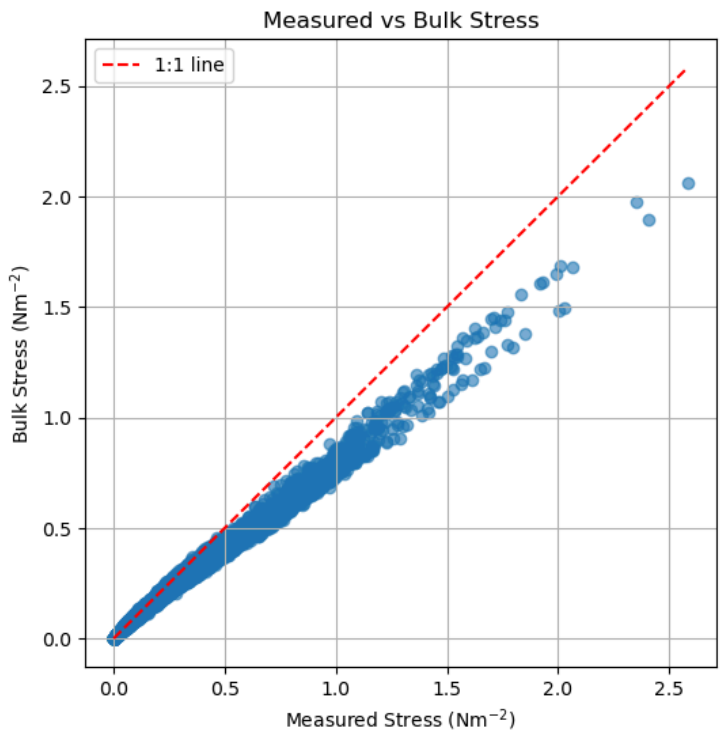
A. A time series of the directly measured buoy data and the calculated bulk estimates for the entire data range is shown below. Quantitatively, the datasets align well: bulk estimates generally capture the variability but exhibit slightly smaller peaks than the measurements. A scatterplot of measured versus bulk values (also shown below) demonstrates a high correlation ($R^2 = 0.999$) and a mean difference of 0.024 N m^{-2}. As expected, bulk estimates tend to underpredict fluxes at the highest wind speeds, highlighting a limitation of bulk parameterizations under extreme conditions.





B. The procedure described in section A was repeated using measured wind stress calculated from absolute wind speed, rather than the relative wind speed previously used in bulk flux calculations. The resulting comparison is shown in the scatterplot below.

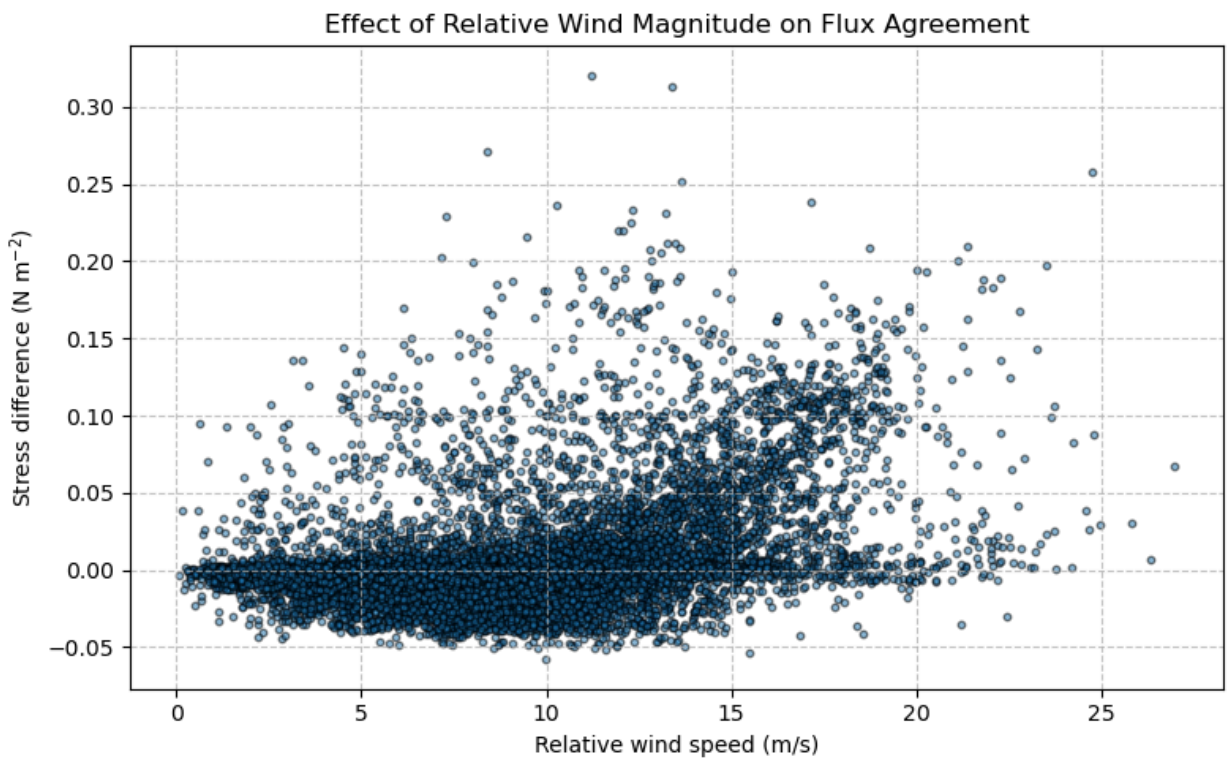




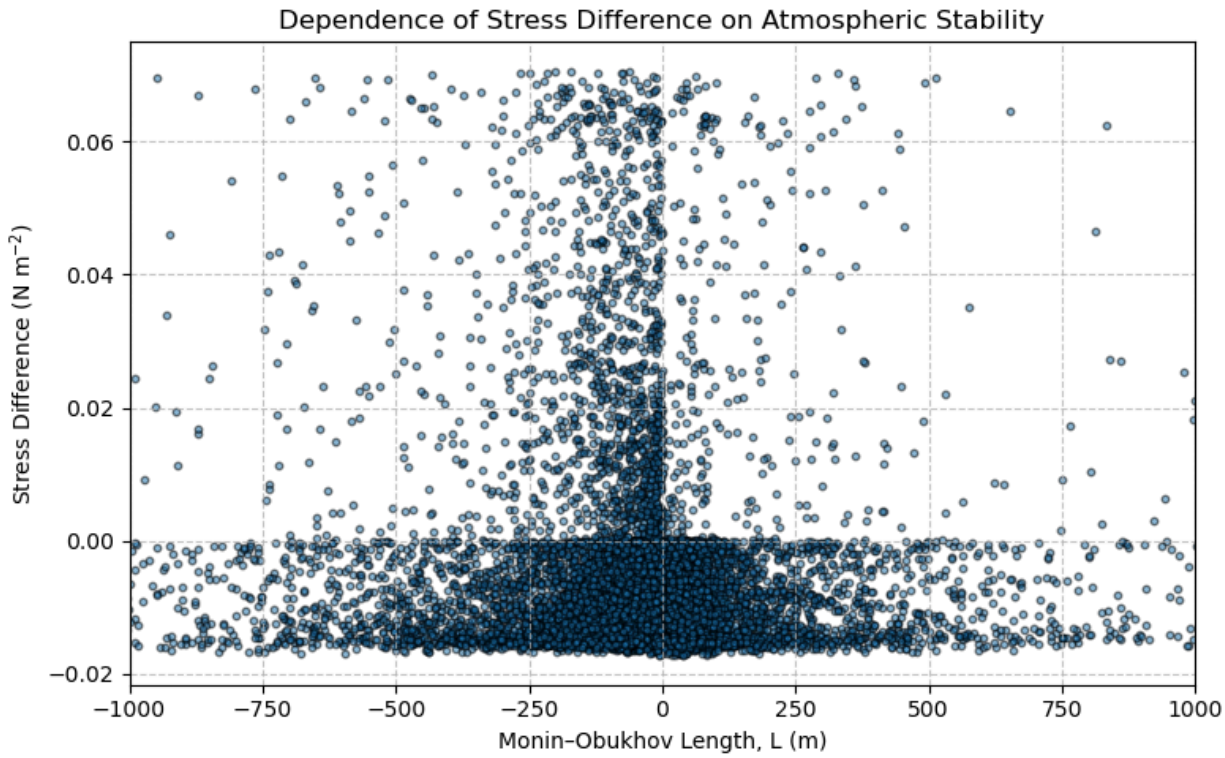
The trends observed with relative wind stress are generally preserved, but bulk calculations now underestimate stress at lower wind speeds, and the scatter in the data is slightly higher ($R^2 = 0.991$, mean difference = 0.064 N m^{-2}). Comparing bulk stress estimates based on relative versus absolute wind speed shows that values computed using absolute wind are consistently lower than those based on relative wind, with the discrepancy increasing at higher wind speeds. This occurs because the average relative wind speed exceeds the absolute wind speed at the Pioneer buoy: along-shelf surface

currents generally oppose the prevailing westerlies, which increases the effective wind speed used in relative-wind-based stress calculations. Since bulk stress scales roughly with the square of wind speed, a small decrease in wind magnitude leads to noticeably lower stress estimates. This explains why bulk calculations based on absolute wind show slightly lower values and greater scatter compared to those based on relative wind.

C. From the scatterplots in B, it is clear that disagreement between estimated and observed values increases with wind speed. Momentum flux agrees best under moderate winds (~5–10 m/s), where the bulk parameterization matches measured stress fairly closely. This is further illustrated in the stress-difference plot below. Estimated values are most accurate at low wind speeds and increasingly underestimate measured stress at high wind speeds (>18 m/s).

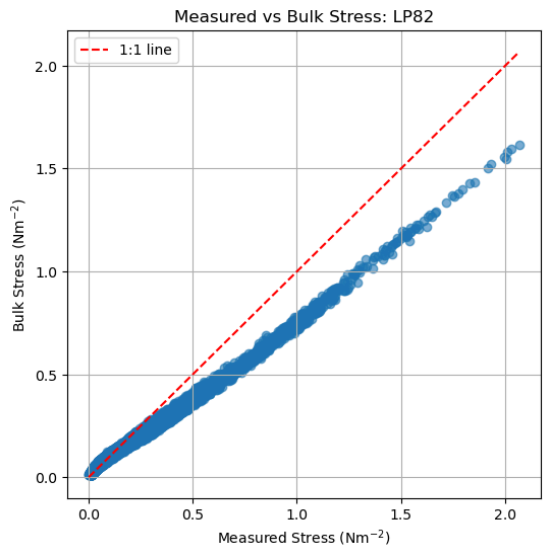


Stability has a secondary effect. In the stress-difference vs. Monin-Obukhov length plot, the most obvious feature is the lack of points in quadrant 1: bulk stress rarely underestimates measured stress under stable conditions ($L > 0$). Under unstable conditions ($L < 0$), disagreement is more variable, consistent with the wider scatter in the plot. Overall, high wind is the primary driver of disagreement, with stability contributing a smaller, systematic effect likely related to the scheme switching estimation regimes depending on stability state.



D. The largest departures from measured stress occur under strong wind conditions, as shown in the time series and scatterplots for the full measurement period. Above ~18 m/s, bulk stress underestimates measured stress, consistent with the ECMWF drag coefficient cap.

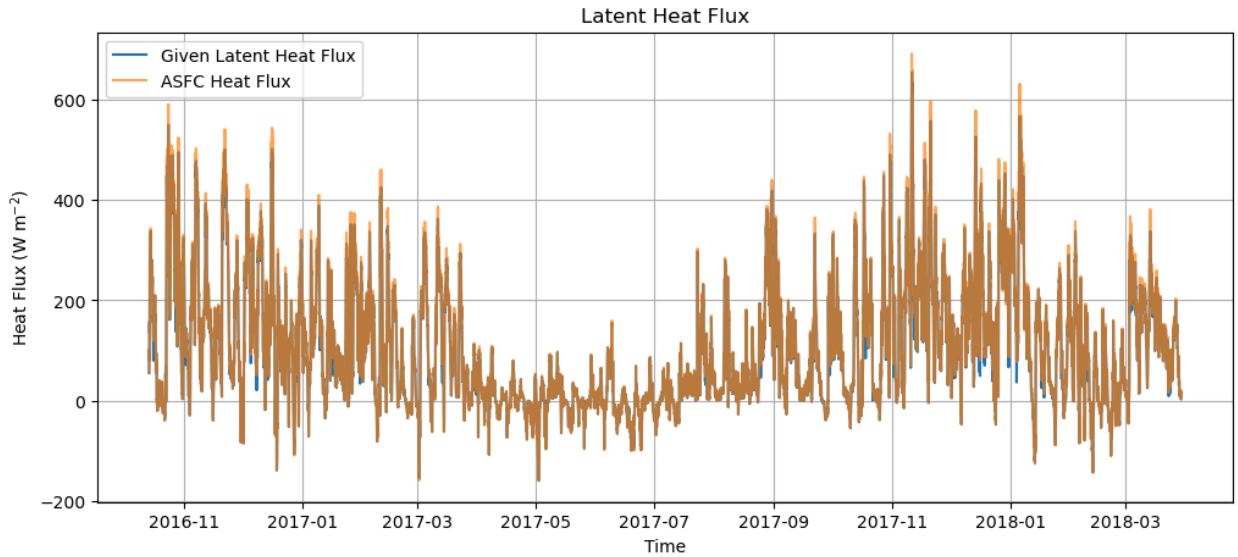
Comparing different schemes highlights some notable differences. LP82 has a high correlation ($R^2 \approx 0.995$), but consistently underestimates stress. The scatterplot below shows disagreement starting above ~0.25 N m^{-2} and reaching nearly 25% at high values.

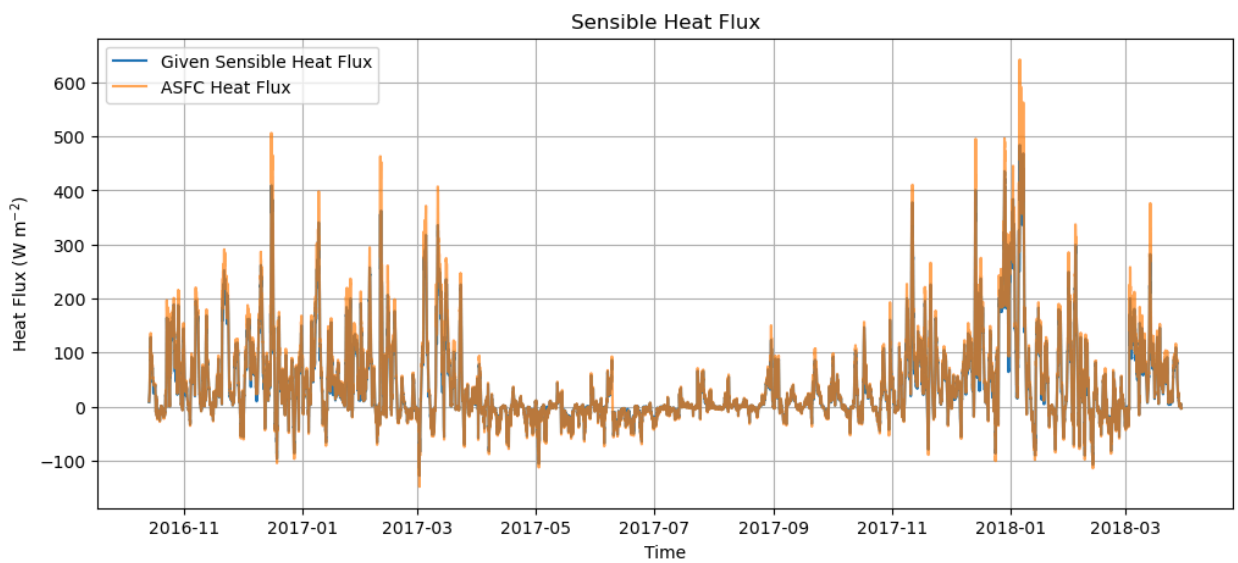


ECMWF performs better overall. Its underpredictions are smaller on average (RMSE = 0.024 N m^{-2} vs. RMSE = 0.078 N m^{-2}), benefiting from wave-dependent roughness, gustiness correction, and stability adjustment, making it generally reliable under the storm-track conditions observed at the Pioneer buoy. That being said, the R^2 values are identical at 0.999. While the drag coefficient cap does slightly limit extreme values, ECMWF provides the most consistent estimates across the full wind range.

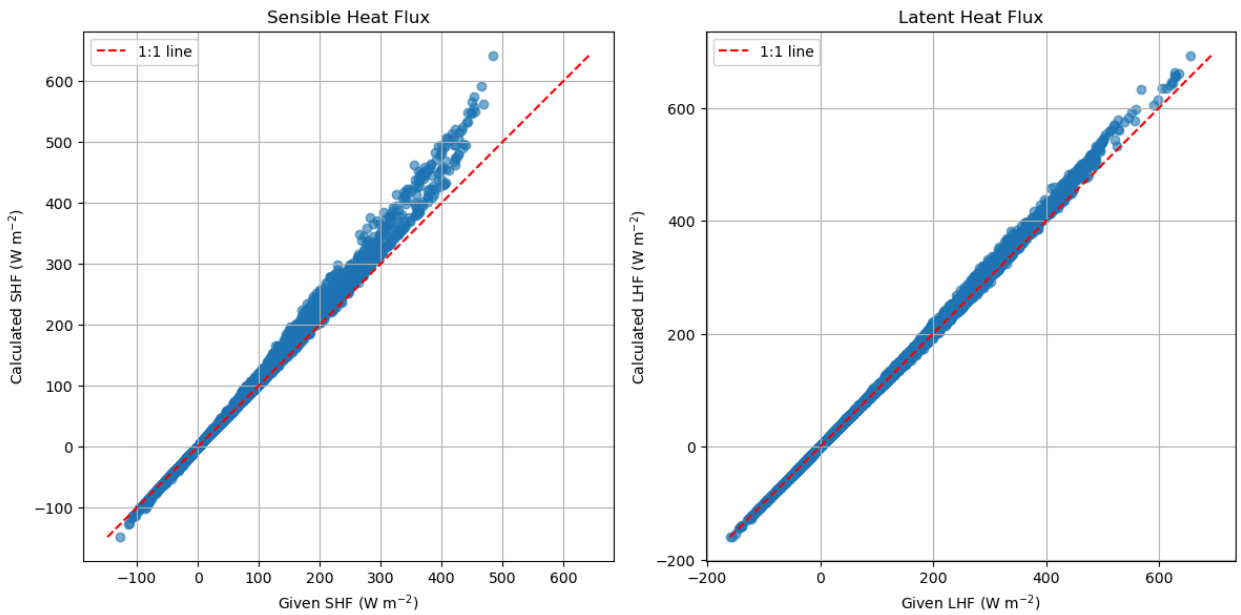
Task 4:

Using the AirSeaFluxCode: latent heat flux, sensible heat flux, heat exchange coefficient (C_t) and moisture exchange coefficient (C_q) were calculated. Time series for both heat fluxes were plotted alongside their given values in the dataset.

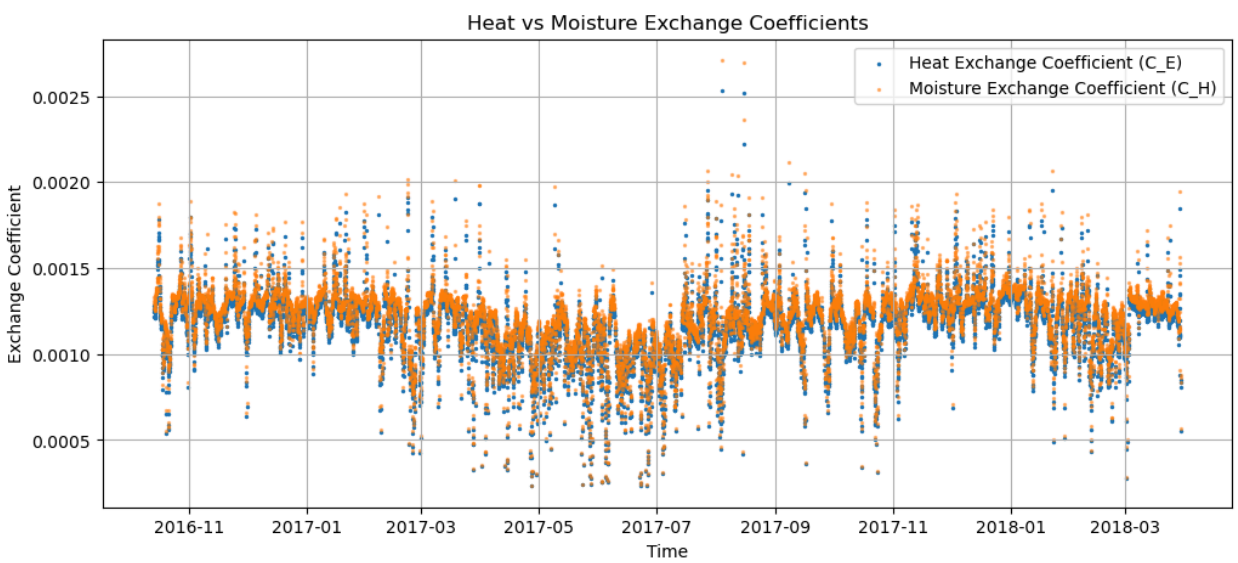




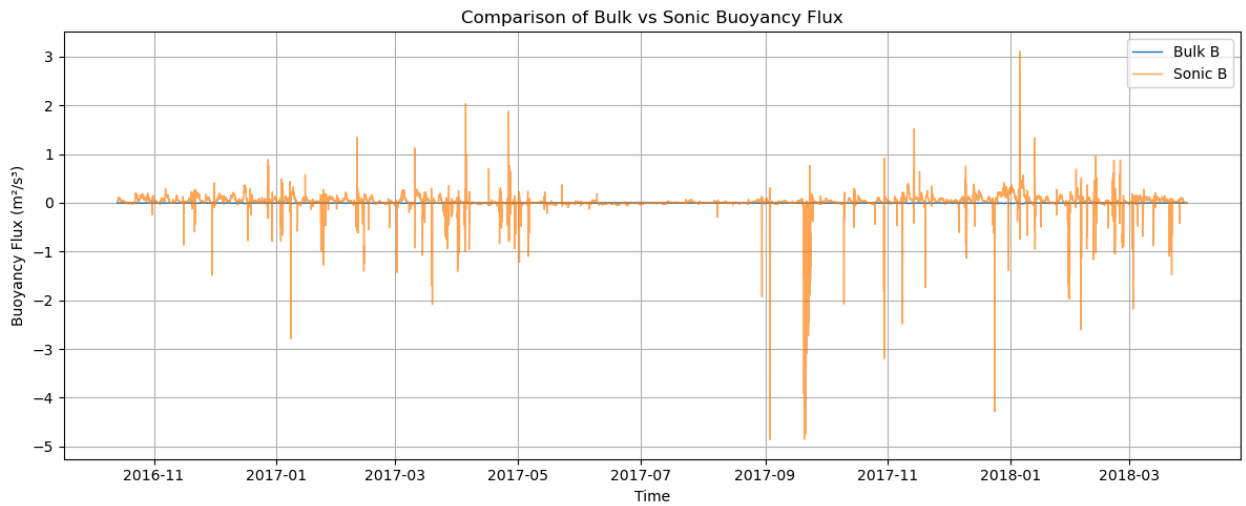
To intercompare the given and calculated values, they were plotted against each other in the below scatterplots. The matchup is very high with an R^2 of 0.995 and RMSE of 11.62 W m^{-2} for sensible, and R^2 of 0.999 and RMSE of 6.45 W m^{-2} for latent. The matchup is unsurprising as the given values were calculated using COARE 3.5, so the differences are due to differences in parameterization schemes, however this does confirm that any bugs in the code have been removed (the discrepancies that were presented in class were fixed by pulling the debugged code on 3/13).



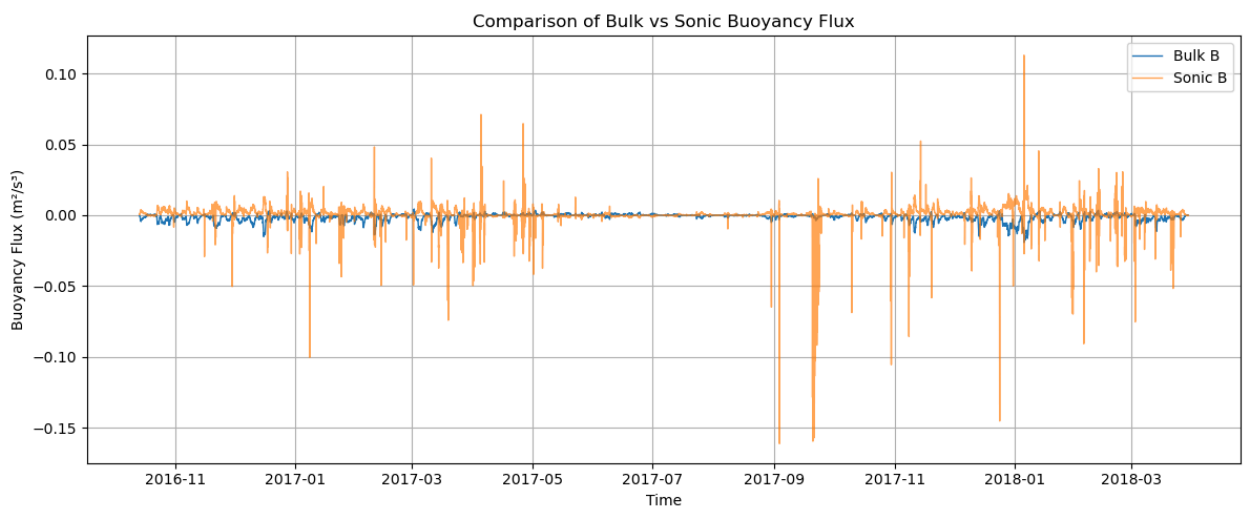
The exchange coefficients were plotted against one another as the given dataset did not provide them, and are very correlated.



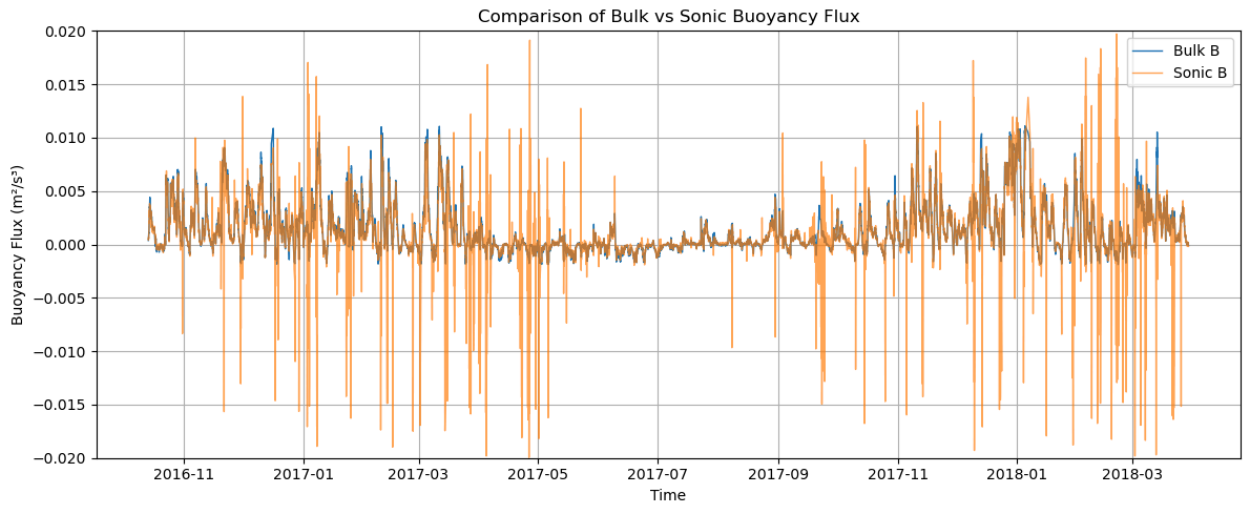
The buoyancy flux was calculated using the equation given in the homework and compared to the 'WTr' variable in the dataset. The resulting time series can be seen below.



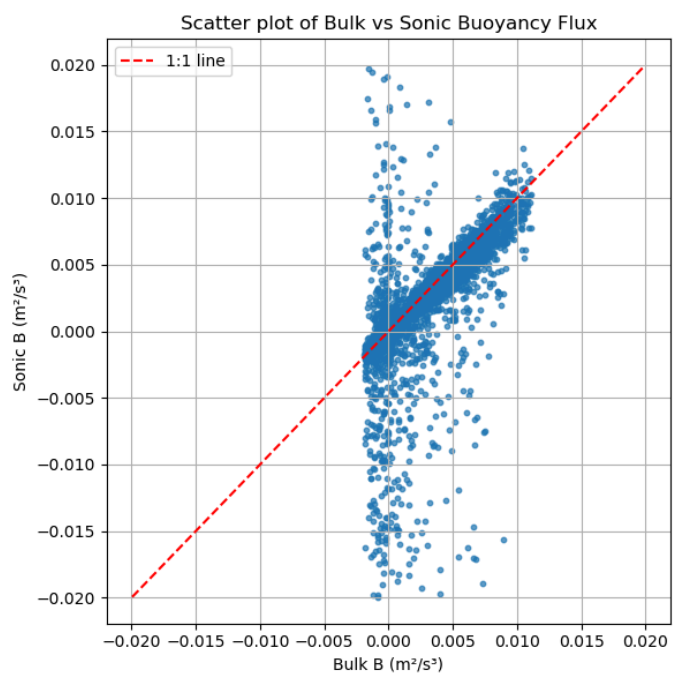
Clearly there is an issue as the measured values are an order of magnitude higher than the measured values. When analyzing the data, it is noted that the units of 'WTr' are m/s*K. It was then theorized that 'WTr' actually corresponds to ω'T', and that the correct representation of buoyancy flux should be $B = g/\theta_v * WTr$. This was plotted and can be seen in the time series below.



While a notable improvement, as the two lines are visible, it is still far from an ideal match. It can be noted that the flux sign convention may be flipped as the R^2 value between the two is -0.292. Additionally, assuming the extreme spikes are from sensor noise, it is possible to filter the unphysical data by removing values that have an absolute value greater than 0.02 $m^2 s^{-2}$. We then clip the resultant top and bottom 1% of values, and get the below time series.



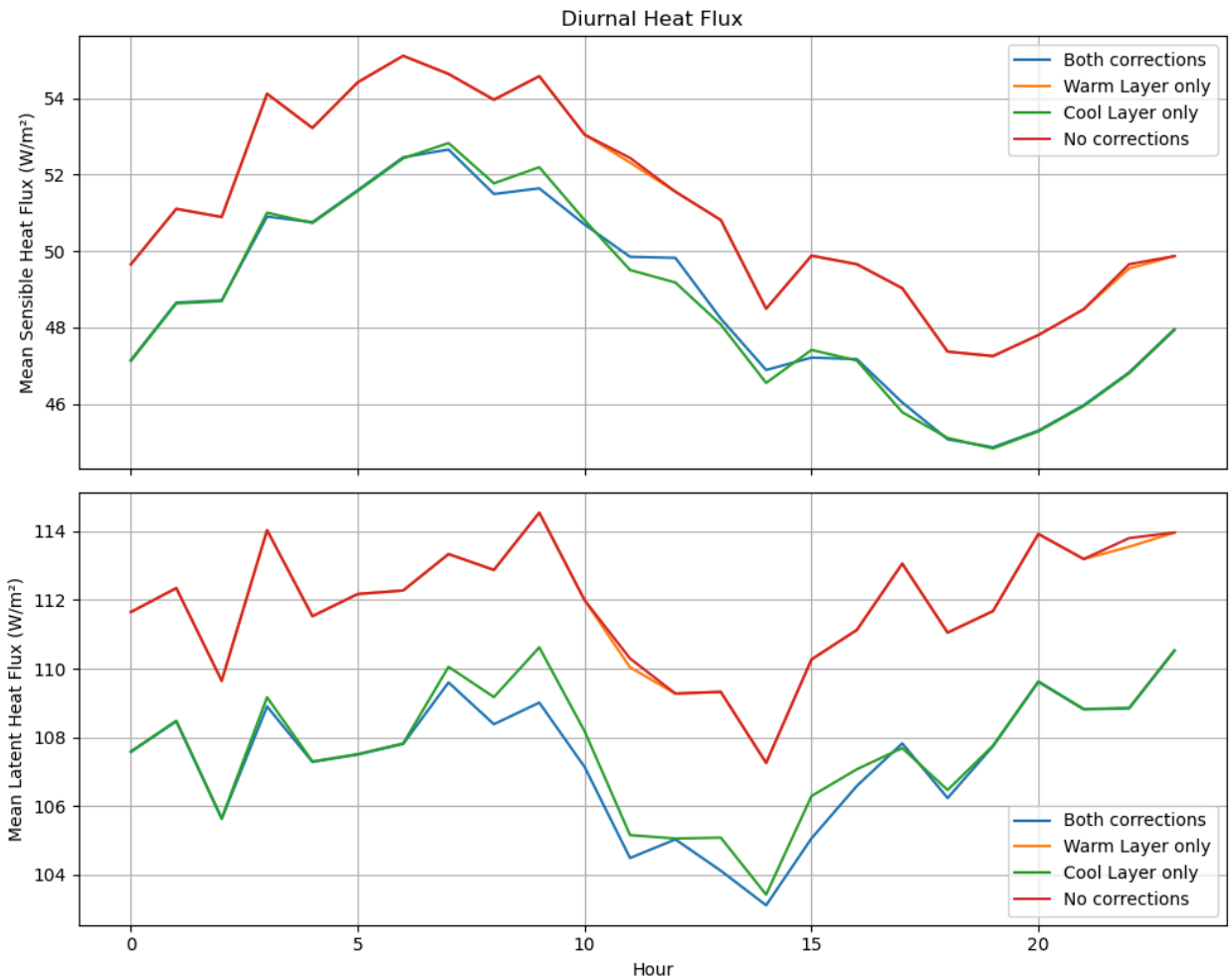
This data, while still noisy, has a far closer correlation to the calculated values. Again plotting a scatter plot intercomparing the two, we find that the filtered data has an R^2 value of 0.695.



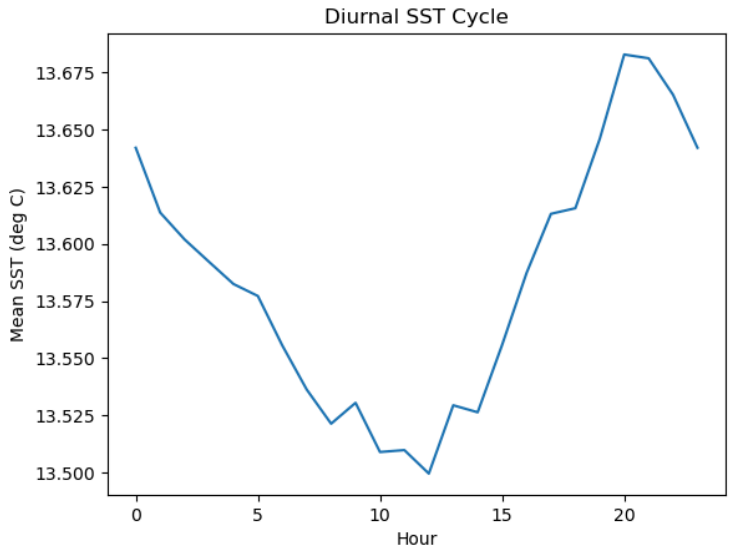
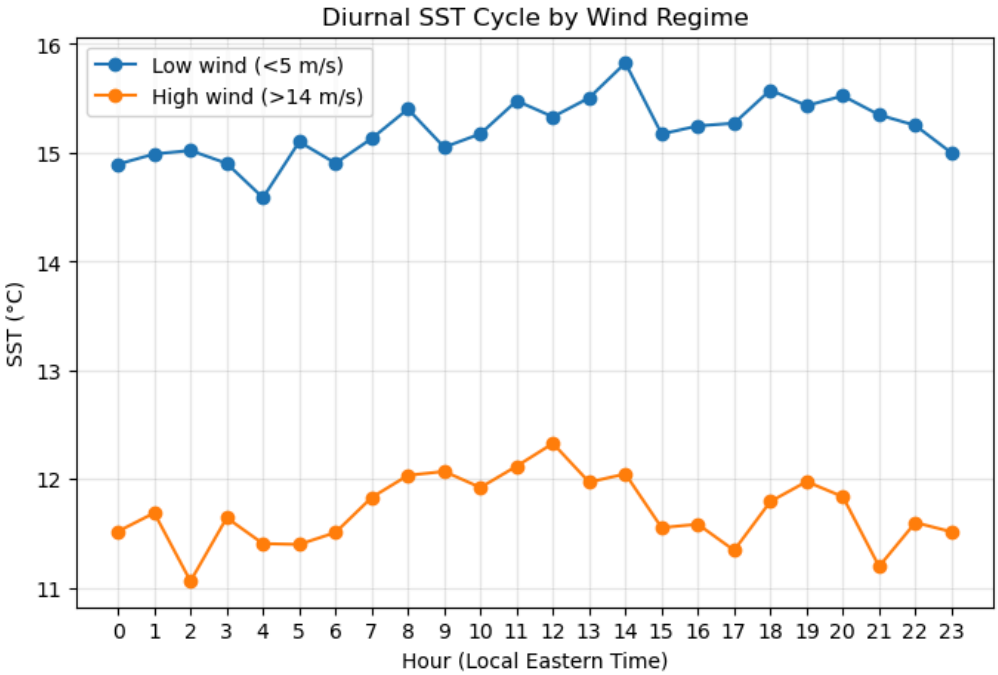
The plot has a strong general trend, with what appear to be clusters of outliers that occur when the measurement is close to 0. The data cleaning undertaken to get these plots is somewhat aggressive, however it does show that there is a correlation and that the sensor may need to be time averaged over a longer period of time, or may be faulty.

Task #5

To analyze the impact of the diurnal warm layer, four runs of the AirSeaFluxCode were performed. Each run toggled the warm- and cool-layer corrections on or off, allowing us to isolate their effects on both latent and sensible heat fluxes. To examine diurnal variability, calculated fluxes were grouped by local hour and plotted in the figure below.



Both flux types show some hourly variation; however, sensible heat flux exhibits a clearer diurnal signal, while latent heat flux varies more irregularly. This is consistent with the expected physical behavior, as sensible heat depends directly on the air-sea temperature difference, whereas latent heat also depends on humidity and wind, which introduce additional variability. In both fluxes, the cool-layer correction dominates, and the warm-layer effect is relatively minor under the conditions sampled.

To examine the influence of wind on diurnal SST, all measurements were binned by local hour and the mean of each group calculated (plot shown below). A pronounced overall diurnal SST

variation is evident. However, this signal becomes less distinct when filtering by wind speed. At the extremes (<5 m/s and >15 m/s), mean SST differs between low- and high-wind conditions, but neither subset exhibits a clear diurnal cycle. The lack of a clear diurnal cycle may be a result of other processes masking the expected diurnal warming.